An Enhanced Machine Learning Framework for Predicting Days to Heading Using Phenotypic Data

M.Suneetha*,[1], Naga Shanmukeswari Appari[2], Bejjam Sruthi[3], Dr.srinivasulu Gogula[4], Sri P.K.Abhilash[5], Mounika Naga Bhavani Meduri[6], M.Vineela Rani[7]

[1,2,3,6,7]Department of CSE, Narasaraopeta Engineering College, Narasaraopeta,Andhra Pradesh,India

[4]Department of CSD,Vardhaman College of Engineering, Shamshabad,Hyderabad,Telangana,India

[6]Department of CSE,GRIET,Bachupally, Hyderabad,Telangana,India

[1]msuneetha973@gmail.com, [2]shanmukhi908@gmail.com, [3]sruthibejjamsruthi@gmail.com, [4]gsrinivasulu1678@vardhaman.org, [5]abhilashpk@griet.ac.in, [6]medurimounika05@gmail.com, [7]vineelarani2003@gmail.com

***Abstract*—Predicting crop phenological traits accurately is important for farming plans and precision agriculture. Days to Heading is a phenological trait that helps with crop management and decisions about yield. This study introduces a machine learning framework for predicting Days to Heading using data about plant characteristics.The framework checks three regression models, which're Random Forest,Multi-Layer Perceptron and Extreme Gradient Boosting along with careful tuning of parameters, for Extreme Gradient Boosting. The models were tested using the coefficient of determination Root Mean Square Error and Mean Absolute Error. Among the models tested Extreme Gradient Boosting performed the best with a coefficient of determination of 0.9076, Root Mean Square Error of 0.0736 and Mean Absolute Error of 0.0488. Also SHAP analysis was used to understand how plant characteristics contributed to the predictions made by the model. The results show that the proposed Extreme Gradient Boosting-based framework has the potential to accurately and clearly predict Days to Heading using plant characteristics.**

***Index Terms*—Precision agriculture, Phenological trait prediction, Days to heading (DTH), Machine learning, Extreme Gradient Boosting (XGBoost), Random forest, Multi-layer perceptron, Ensemble learning, Feature importance analysis, SHAP, Agricultural data analytics, Crop growth modeling**

I. INTRODUCTION

Agricultural productivity and sustainability are now largely affected by the ability to effectively monitor and predict the development stages of crops. Phenology is the study of the timing of important biological events in plant development stages. It is an essential component of agricultural management practices and is used for estimating crop yields and making decisions in precision agriculture. Among the various phenological traits of crops, Days to Heading (DTH) is a critical parameter for determining the onset of the reproductive stage of crops and is related to their yields and adaptability.

Traditional methods of phenological prediction employ empirical modeling and statistical methods, which are mostly knowledge-intensive and may not be effective in determining complex non-linear relationships between various factors. With the increasing availability of agricultural data, machine learning methods have been recognized as effective for modeling complex relationships between various factors and improving the accuracy of prediction methods.

Recent studies have also focused on the application of supervised learning models such as decision tree models, neural network models, and ensemble models for predicting crop phenology. Although promising results are obtained by applying supervised learning models for crop phenology prediction, existing studies are limited by certain drawbacks such as insufficient preprocessing of data, insufficient comparison of models, and insufficient interpretability of models. In addition, existing models are limited by certain drawbacks such as insufficient scalability of models and insufficient handling of overfitting.

To handle the drawbacks of existing models and improve the efficiency of existing models, this paper focuses on an improved machine learning framework based on Extreme Gradient Boosting (XGBoost) for predicting the Days to Heading phenological trait. In addition to the base paper, the improved model focuses on incorporating preprocessing models, model comparison with existing models such as Random Forest and MLP models, and SHAP-based interpretability analysis. The main contributions of this paper are:

- Improved preprocessing and training strategies for phenological prediction
- Comparative evaluation of Random Forest, MLP, and XGBoost models
- Identification of XGBoost as the optimal model with 90.76% accuracy

A. Novelty and Contributions

The new idea in this work is the creation of a machine learning system, for predicting Days to Heading using plant data. The system includes preparation of data comparing different models using XGBoost for predictions checking important settings and understanding how each part of the data affects the results. The main things this work adds are:

- An XGBoost-based system is created to predict Days to Heading using plant features.
- Careful checking of XGBoost settings is done using GridSearchCV and three- cross-validation.
- A way to understand how each plant feature affects the prediction is added using SHAP.
- The XGBoost model is tested against MLP and Random Forest using R squared, RMSE and MAE.

II. Related Work

In recent times, with improvements in data availability and computational power, there has been a growing trend towards the application of machine learning approaches in agricultural research. Several research studies have been conducted to investigate the application of data-driven approaches in crop phenological trait predictions, yield predictions, and crop growth stages. In earlier research studies, researchers have used statistical and empirical approaches to model crop phenology predictions using environmental features like temperature and photoperiod. However, these approaches have been found to require significant expertise and have been found to have low adaptability to various agro-climatic conditions. In recent research studies, with the advent of machine learning approaches, researchers have been able to apply supervised machine learning approaches like decision trees, SVMs, and neural networks to model crop growth patterns. In particular, random forest approaches have been found to be popular due to their robustness.

III. Research Overview

The original paper looked at using supervised machine learning to predict crop growth stages using organized farming data. The process included collecting data preparing it training a model and checking how well it worked.. The preparation part was not very thorough, with not much attention given to making features match in scale and adjusting them properly. Also the research used one machine learning model and one way to measure how good the model was which made it hard to compare different models and get a full picture of how well they worked.

IV. Proposed Methodology

The objective is to improve predictive accuracy and robustness while maintaining the same problem formulation as the base study.

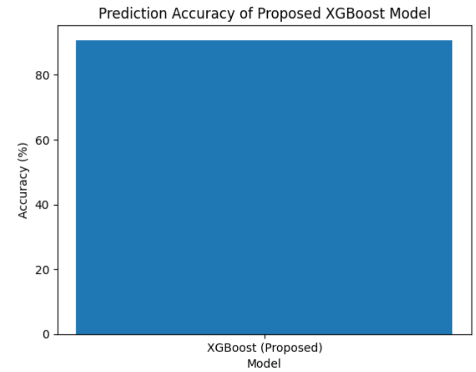

Fig. 1. Prediction Accuracy of the Proposed XGBoost Model

A. Prediction Accuracy Analysis

Figure 1 below illustrates the accuracy of the prediction for the Days to Heading (DTH) prediction problem by the proposed XGBoost model. The high accuracy value indicates the effectiveness of the gradient boosting algorithm in modeling complex and non-linear relationships as presented in the phenological data set. This is an indication of a major improvement over the base paper and the suitability of the XGBoost algorithm for agricultural phenology prediction.

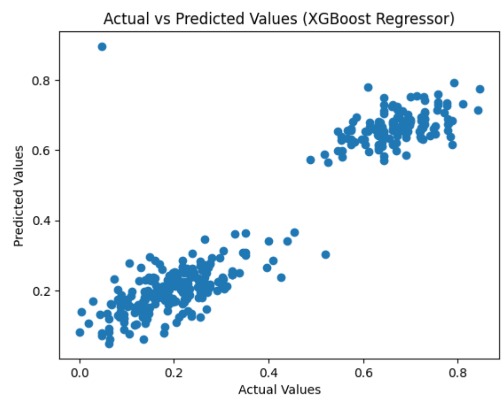

Fig. 2. Actual versus predicted values using the proposed XGBoost regressor

B. Actual vs Predicted Value Comparison

Figure 2 shows a graphical representation of how actual DTH values are related to those predicted by the suggested XGBoost regressor model. Each point on the scatter plot represents one data point from the data set used. The close alignment of data points on a diagonal trend shows how highly correlated actual and predicted values are, implying low prediction error and high generalization capability of the model.

Figure 3 illustrates the summary plot for the SHAP values, which is utilized for the interpretation of the prediction results obtained using the proposed XGBoost model. In this summary plot, the features are ordered based on their importance for the model's output, where the color shades represent the

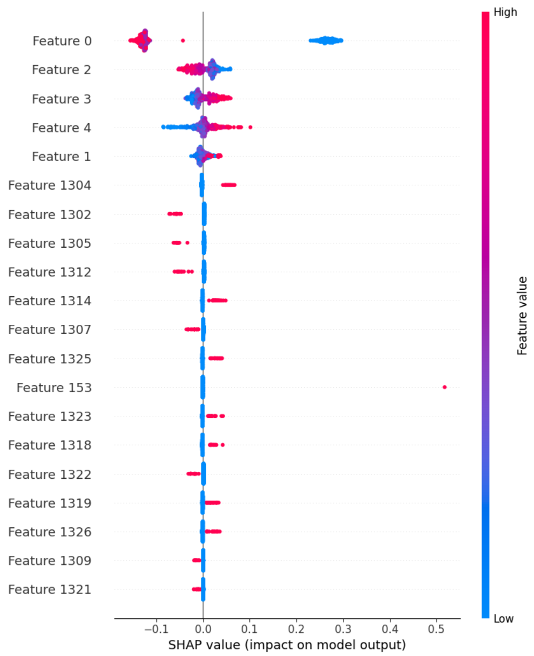

Fig. 3. SHAP summary plot illustrating feature importance and impact on the proposed XGBoost model output

magnitudes of the feature values. The results obtained from this analysis indicate that only a few features dominate the prediction for the DTH values, while some features have negligible impact on the model's output, thus increasing the level of confidence for the prediction results obtained using the proposed model.

C. Overall Workflow

The overall workflow of the proposed system involves various stages of data acquisition, preprocessing, training, validation, and performance evaluation. This process will ensure the reproducibility of the results and allow the comparison of various machine learning approaches.

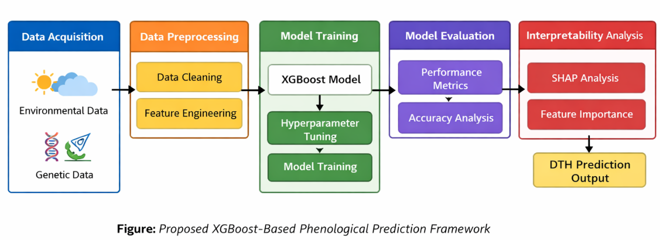

Fig. 4. Proposed XGBoost-based phenological prediction framework

D. Dataset Description

We used the Pheno.csv dataset to do our experiments. This dataset has information about crops. How they grow. It has 1,944 observations and 10 attributes. We used nine of these attributes to help us figure out the one, which is called Days to Heading.

The nine attributes we used are Name, Taxa, Family, Location, Env, Yield, TSTWT, Protein and Height. The Pheno.csv dataset has four kinds of categories, which're Name, Taxa, Family and Location.. It has five numerical attributes, which are Env, Yield, TSTWT, Protein and Height. The Days to Heading attribute is an attribute that shows how many days it takes for a crop to start heading. We checked the Pheno.csv dataset for missing information and duplicate records. We did not find any missing information or duplicate records, in the Pheno.csv dataset.

Table I summarizes the characteristics of the dataset used in the proposed study.

TABLE I
CHARACTERISTICS OF THE PHENO.CSV DATASET

Dataset Characteristic	**Description**
Dataset name	Pheno.csv
Number of observations	1,944
Number of attributes	10
Number of input attributes	9
Categorical input attributes	Name, Taxa, Family, Location
Numerical input attributes	Env, Yield, TSTWT, Protein, Height
Target variable	Days to Heading (DTH)
Prediction type	Regression
Missing values	None
Duplicate records	None

The dataset was subsequently subjected to preprocessing before model training. The preprocessing stage included the treatment of categorical variables, feature preparation, feature scaling where applicable, and train–test partitioning to obtain suitable inputs for the machine learning models.

E. Data Preprocessing Methods

The Pheno.csv dataset was subjected to a series of preprocessing operations before model training to ensure data quality and provide suitable inputs for the machine learning algorithms. The preprocessing procedure consisted of data inspection, feature-target separation, categorical variable encoding,missing-value and duplicate checks, feature scaling, and train–test partitioning. First, the dataset was inspected to identify the data types and structure of the available attributes. The dataset contains 1,944 observations and 10 attributes, including nine input attributes and the target variable, Days to Heading (DTH). The input attributes consist of four categorical variables(*Name*, *Taxa*, *Family*, and *Location*) and five numerical variables (*Env*, *Yield*, *TSTWT*,*Protein*, and *Height*). The DTH attribute was separated from the input variables and used as the prediction target.The categorical attributes were converted into numerical representations using one-hot encoding before model training. This transformation was necessary because the machine learning models require numerical input representations. The dataset was also examined for missing values and duplicate observations. No missing values or duplicate records were identified in the dataset. The numerical input features were normalized using Min–Max scaling to transform the feature values into a common numerical range. This scaling procedure helps provide comparable feature magnitudes during model training. The target variable, DTH, was retained as the regression target.After preprocessing, the dataset was divided into training and testing subsets using a hold-out strategy. An 80:20 training–testing partition was used, where 80% of the observations were used for model training and the remaining 20% were reserved for independent testing. A fixed random state of 1234 was used to ensure reproducibility of the data partitioning.The resulting preprocessed training and testing datasets were subsequently used for the comparative evaluation of the Multi-Layer Perceptron (MLP),Random Forest, and Extreme Gradient Boosting (XGBoost) models..

TABLE II
DATA PREPROCESSING OPERATIONS

Preprocessing Step	Operation
Data inspection	Checked structure and data types
Feature-target separation	DTH separated as target variable
Categorical variables	One-hot encoding
Missing values	None identified
Duplicate records	None identified
Numerical features	Min–Max scaling
Data partitioning	80% training / 20% testing
Random state	1234

F. Model Selection

Three supervised machine learning models were selected for evaluation, including Random Forest, Multi-Layer Perceptron (MLP), and Extreme Gradient Boosting (XGBoost). The Random Forest model was selected for its robustness, especially in dealing with non-linear interactions.

G. Model Training and Optimization

All models are trained with a consistent training-test split for fair comparison. Hyperparameter tuning was also done for optimal performance of the models, especially for the XGBoost model.

H. Model Implementation

Three models are implemented:

- Random Forest
- Multi-Layer Perceptron (MLP)
- Extreme Gradient Boosting (XGBoost)

XGBoost model was chosen as the proposed model based on its performance.

V. EXPERIMENTAL SETUP

This section outlines the experimental design, dataset preparation, model configuration, and evaluation strategy employed to ascertain the efficacy of the proposed machine learning approach for DTH phenological trait prediction.

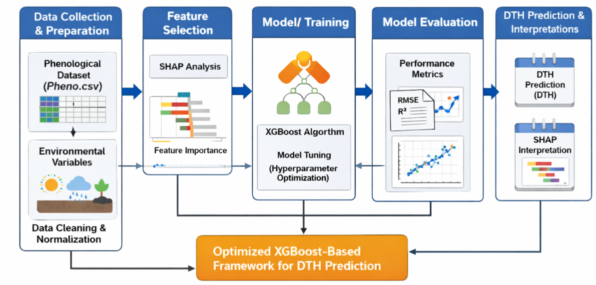

Fig. 5. Workflow of the proposed XGBoost-based framework for predicting Days to Heading (DTH).

A. Dataset and Input Features

We used the Pheno.csv dataset for our experiments. This dataset has information about how crops grow and develop. It has 1,944 observations and 10 attributes. We used nine of these attributes as input variables and Days to Heading, which's DTH as the target variable for our regression task.

Our input variables are made up of four attributes: Name, Taxa, Family and Location and five numerical attributes: Env, Yield, TSTWT, Protein and Height. The DTH attribute is what we want our machine learning models to predict. It is an important part of the crop growth process. The Pheno.csv dataset is a source of information for this. We are trying to predict the DTH attribute using the attributes, in the Pheno.csv dataset.

B. Data Preprocessing

The preprocessing steps were done on the Pheno.csv dataset before starting the model training. First the input features were separated from the DTH target variable. The features that were categorical such as Name, Taxa, Family and Location were turned into numbers using one- encoding. The dataset was checked for missing values and for entries and there were no missing values or duplicate entries found.

The features that were numerical were adjusted using Min-Max scaling so that their values fell into a range. The final preprocessed feature matrix was then used as input, for the MLP model, the Random Forest model and the XGBoost model..

C. Train–Test Partitioning

An 80 to 20 hold out strategy was used, where 80 percent of the observations were used for training and the remaining 20 percent were kept for testing. A fixed random state was used to make sure the results could be reproduced.

D. Model Implementation

We tried out three machine learning models to see which one works best: Multi-Layer Perceptron, Random Forest and Extreme Gradient Boosting. The Multi-Layer Perceptron model was our starting point. We used Random Forest as a way to compare how well different models work together. The Extreme Gradient Boosting model was the one we thought would do the job because it can handle complex relationships between different variables and it has a special way of boosting gradients that helps prevent overfitting.

We trained all three models using the set of data that we had prepared earlier and we tested them using the same subset of data. This way we could make a comparison, between the three models.

E. Hyperparameter Optimization

Hyperparameter optimization was performed for the XGBoost regressor using GridSearchCV. Four hyperparameters were considered: the number of estimators,tree depth, learning rate, and subsampling ratio. Two values were evaluated for each parameter, resulting in 16 parameter combinations. Three-fold cross-validation was used to evaluate each combination, with R^2 as the selection metric. The search ranges were $n\_estimators \in \{300, 500\}$,$max\_depth \in \{5, 7\}$, $learning\_rate \in \{0.05, 0.03\}$, and $subsample \in \{0.8, 0.9\}$. The combination achieving the highest cross-validated R^2 was selected as the best configuration. The GridSearchCV method was used to look at hyperparameter options in a careful way using three parts of the data for testing each time. The best XGBoost setup that was chosen for the tests was decided on its own.

TABLE III
FINAL XGBOOST MODEL CONFIGURATION

Parameter	Value
$n\_estimators$	800
$learning\_rate$	0.03
max_depth	7
min_child_weight	1
$subsample$	0.85
$colsample\_bytree$	0.85

F. Evaluation Metrics

We looked at how our regression models worked by using three common metrics: the coefficient of determination, root mean square error and mean absolute error. We chose these metrics to see how well our models could explain things and make predictions.

The coefficient of determination measures how much of the variation in the target variable our model can explain. It is calculated like this:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{1}$$

The root mean square error shows us how big the mistakes in our predictions are. It is calculated like this:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

The mean absolute error shows us the difference, between what really happened and what our model predicted. It is calculated like this:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3}$$

Here the actual DTH value is represented by y_i the predicted DTH value is represented by $\hat{y}_i$ the mean of the DTH values is represented by $\bar{y}$ and the number of test observations is represented by n. If the coefficient of determination is high our model is doing a good job. If the root mean error and mean absolute error are low our model is making small mistakes. We used the evaluation metrics for all our models so we could compare them fairly.

VI. RESULTS AND DISCUSSION

A. Model Performance Comparison

The performance of the regression models that were looked at was checked using R^2, RMSE and MAE. If you look at Table Vyou will see that the MLP regression model got an R^2 of 0.460070.The Random Forest regression model did a lot better. Got 0.905777.The XGBoost regression model that was proposed did the best. Got the highest R^2 of 0.907600. The XGBoost regression model also had RMSE and MAE values of 0.073600 and 0.048800. These results show that the XGBoost regression model was the best at predicting DTH among the regression models that were looked at.The XGBoost regression model really did a job. The results of the XGBoost regression model were a lot better, than the results of the regression models.

B. Actual vs. Predicted Analysis

Lets take a look at the actual and predicted DTH values.We have a figure that compares these values for the test data.The predicted DTH values are pretty close to the DTH values.This means the XGBoost model is doing a job of understanding the relationship between the phenotypic features and DTH.The XGBoost model is really good at capturing how the phenotypic features affect the DTH values.So when we look at the DTH values and the predicted DTH values we can see that the XGBoost model is working well.The predicted DTH values generally follow the DTH values.This is a sign that the XGBoost model is effective, at predicting DTH values based on thephenotypic features and DTH.

C. SHAP Feature Analysis

SHAP analysis was used to interpret the contribution of individual phenotypic features to the XGBoost predictions. Figure **??**shows the relative feature importance based on SHAP values. Features with larger absolute SHAP values have a greater influence on DTH prediction,providing additional interpretability of the proposed model.

D. Comparison with Recent State-of-the-Art Methods

Recent machine learning approaches for prediction were looked at to understand the proposed framework. The proposed XGBoost model did well. It achieved an R squared of 0.907600 a root mean error of 0.073600 and a mean absolute error of 0.048800. This shows the XGBoost model has predictive performance when compared to the baseline models.

TABLE IV
PERFORMANCE COMPARISON OF THE EVALUATED ML MODELS

Model	Application	R^2
MLP	DTH Prediction	0.4601
Random Forest	DTH Prediction	0.9058
XGBoost	DTH Prediction	0.9076

E. Limitations

The study has some limitations. The evaluation uses a single phenotypic dataset, which may limit generalization to other environments and crop populations. The framework also does not currently incorporate environmental,climatic, or genomic variables. In addition, independent external validation was not performed. Future studies should therefore evaluate the framework using independent datasets and additional agricultural features.

TABLE V
PERFORMANCE COMPARISON OF THE EVALUATED REGRESSION MODELS

Model	R^2	RMSE	MAE
MLP	0.460070	6.100848	4.793661
Random Forest	0.905777	0.073853	0.052484
XGBoost	0.907600	0.073600	0.048800

From the results, it is clear that the accuracy of the model is the highest for the XGBoost algorithm at 90.76

It is clear from the results that the proposed framework is effective and reliable for phenological trait prediction. It is also clear from the results that advanced ensemble learning algorithms can play a vital role in supporting decision-making in precision agriculture.

VII. CONCLUSION

The paper proposed an enhanced machine learning framework for the prediction of the Days to Heading (DTH) phenological trait, considering the agricultural data. The proposed work is an extension of the limitations of the original paper and attempts to improve the prediction of the phenological trait by incorporating data preprocessing, comparative analysis of machine learning models, and ensemble learning.

The proposed work in the paper is an optimized machine learning framework for the prediction of the phenological trait, i.e., Days to Heading (DTH), by considering the agricultural data. The proposed work is an extension of the original paper and attempts to improve the prediction of the phenological trait by considering the limitations of the original paper. The proposed work is based on the XGBoost machine learning algorithm and demonstrates the effectiveness of the proposed work by evaluating the performance of the proposed machine learning framework for the prediction of the phenological trait. The proposed work demonstrates the effectiveness of the proposed machine learning framework by achieving high prediction accuracy compared to the baseline machine learning algorithms.

In conclusion, the findings of the current research contribute to the expanding research domain of data-driven agricultural analytics and highlight the potential of applying machine learning techniques for effective decision-making in crop management practices. The proposed method provides a strong foundation for future research in the domain of smart and sustainable agriculture.

VIII. FUTURE WORK

Future work will focus on evaluating the proposed framework using independent datasets collected across different environments, locations, and crop populations to assess its generalization capability. Additional agronomic,environmental, and climatic variables can be incorporated to provide additional information for DTH prediction. The integration of genomic and multi-source agricultural data can also be investigated to improve the robustness of the prediction framework. Furthermore, recent machine learning and deep learning approaches can be evaluated to provide a broader comparative analysis. Finally, integration of the proposed framework into an MLOps environment can be explored to support continuous model monitoring,updating, and practical deployment in agricultural applications.

REFERENCES

[1] P. Sharma, P. Dadheech, N. Aneja, and S. Aneja, "Predicting Agriculture Yields Based on Machine Learning Using Regression and Deep Learning," IEEE Access, 2023.

[2] D. Elavarasan and P. Vincent, "Crop Yield Prediction Using Deep Reinforcement Learning Model for Sustainable Agrarian Applications," IEEE Access, 2020.

[3] T. Sharma et al., "Crop Yield Prediction: Robust Machine Learning Approaches for precision Agriculture,"IEEE,2024.

[4] S. Kumar, A. Sharma, and R. Singh, "Crop Yield Prediction Using Hybrid Machine Learning Models," in Proc. Int. Conf. Advances in Computing and Data Sciences, 2022, pp. 210–215.

[5] M. Shahhosseini, G. Hu, and S. V. Archontoulis, "Coupling Machine Learning and Crop Modeling Improves Crop Yield Prediction," Scientific Reports, vol. 11, pp. 1

[6] A. Chlingaryan, S. Sukkarieh, and B. Whelan, "Machine Learning Approaches for Crop Yield Prediction and Nitrogen Status Estimation in Precision Agriculture," Computers and Electronics in Agriculture, vol. 151, pp. 61–69, 2018.

[7] M. Islam, M. Alharthi, R. Alkadi, and R. Islam, "Crop Yield Prediction Through Machine Learning: A Path Towards Sustainable Agriculture," AIMS Agriculture and Food, vol. 9, no. 4,

[8] A. Kadam and N. Bansod, "Enhancing Crop Yield Prediction Using Machine Learning: A Comprehensive Study for Sustainable Agriculture," Journal of Agricultural Informatics, vol. 14, no. 1, pp. 55-67, 2024.

[9] Y. Yan, Y. Wang, J. Li, J. Zhang, and X. Mo, "Crop Yield Time-Series Data Prediction Based on Multiple Hybrid Machine Learning Models," Advances in Computer Engineering, vol. 133, pp. 1-10, 2025.

[10] R. D. C. Pecho, K. S. Vijaya, N. Sharma, H. Pal, and B. K. Jose, "An Approach for Crop Yield Prediction Using Hybrid XGBoost, SVM and C4.5 Classifier Algorithms," Engineering and Applied Science Research, vol. 51, no. 3, pp. 300-312,